

TRANSIT GATEWAY PEERING

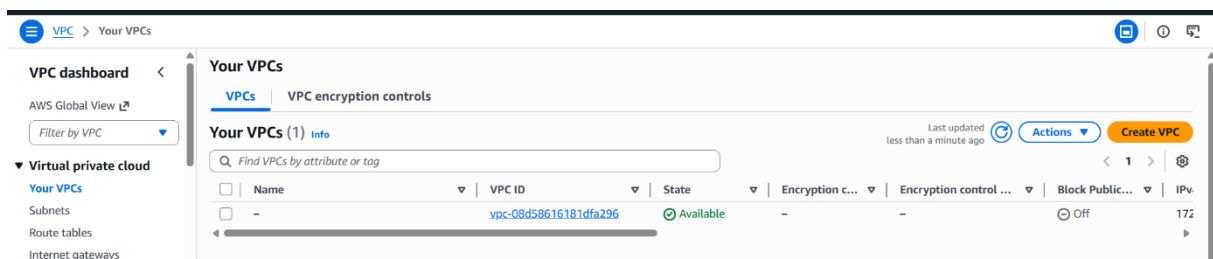
Definition

Transit Gateway Peering is used to connect Transit Gateways in different regions. It allows VPCs in different regions to communicate through their Transit Gateways.

HANDS-ON FOR TRANSIT GATEWAY PEERING

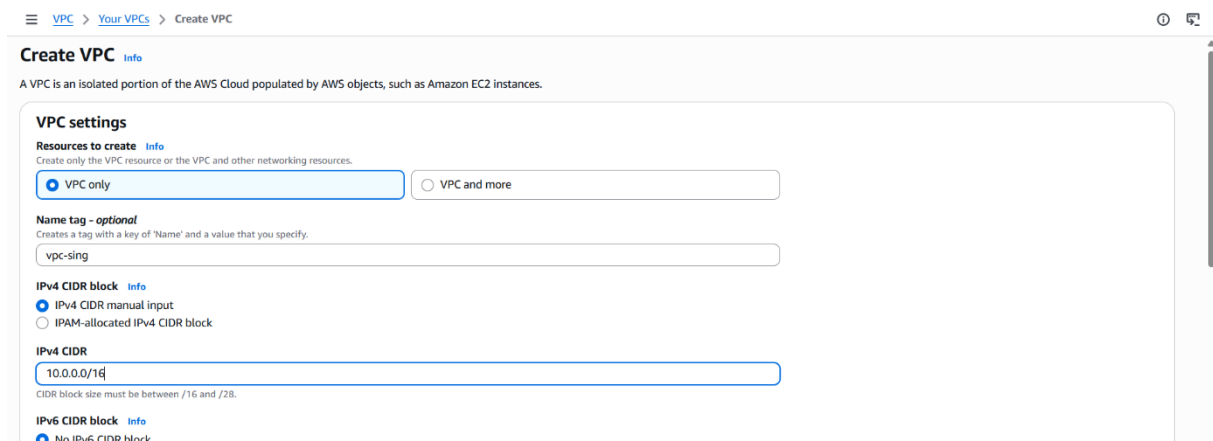
Screenshot (Singapore)

1. Go to VPC
2. Click Create



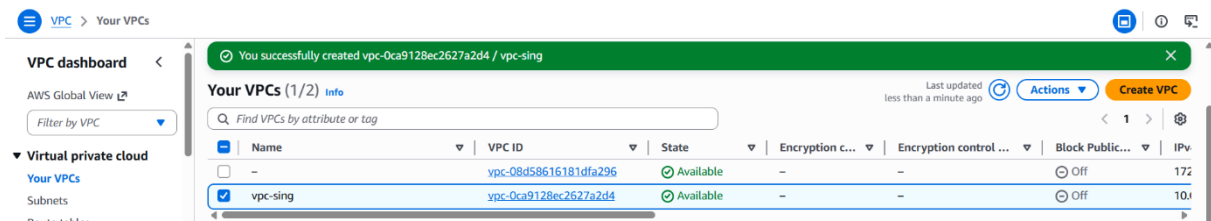
Screenshot: Create VPC (Singapore)

1. Enter VPC name
2. Set CIDR to 10.0.0.0/16
3. Click Create VPC



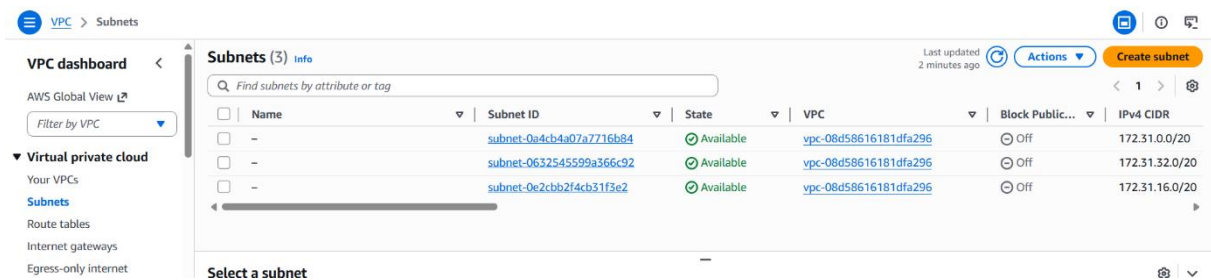
Screenshot: VPC Created (Singapore)

1. View VPC
2. VPC created successfully



Screenshot: Create Subnet (Singapore)

1. Go to Subnets
2. Click Create subnet



Screenshot: Create Subnet (Singapore)

1. Select VPC
2. Enter Subnet name
3. Select Availability Zone
4. Set CIDR to 10.0.1.0/24
5. Click Create subnet

VPC
Subnets
Create subnet

Create subnet

VPC

VPC ID

Create subnets in this VPC.

vpc-0ca9128ec2627a2d4 (vpc-sing)

vpc-08d58616181dfa296
172.31.0.0/16 (default)

vpc-0ca9128ec2627a2d4 (vpc-sing)
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

sub-sing

The name can be up to 256 characters long.

Availability Zone

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Singapore) / apse1-az2 (ap-southeast-1a)

IPv4 VPC CIDR block

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

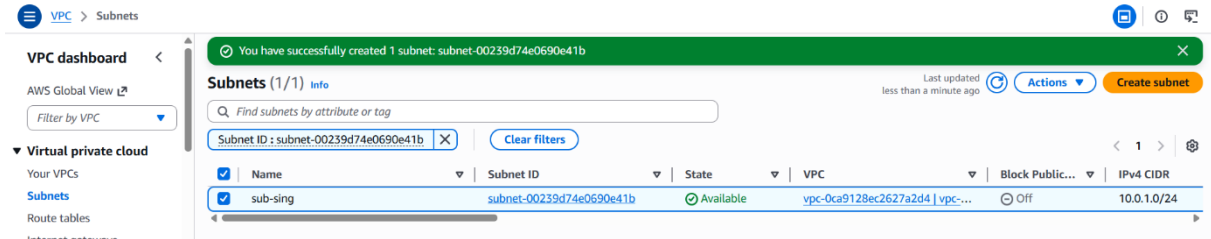
10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24 256 IPs

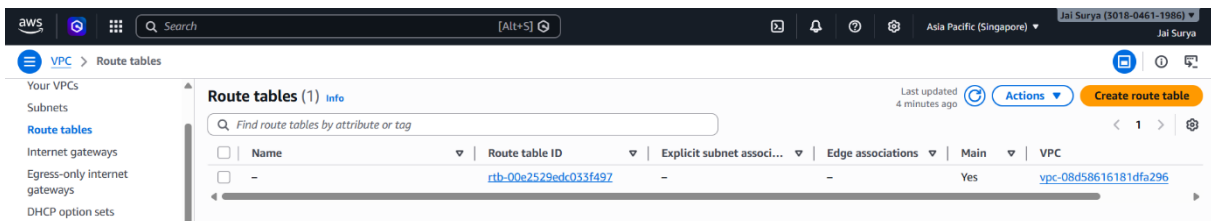
Screenshot: Subnet Created (Singapore)

1. View Subnet
2. Subnet created successfully



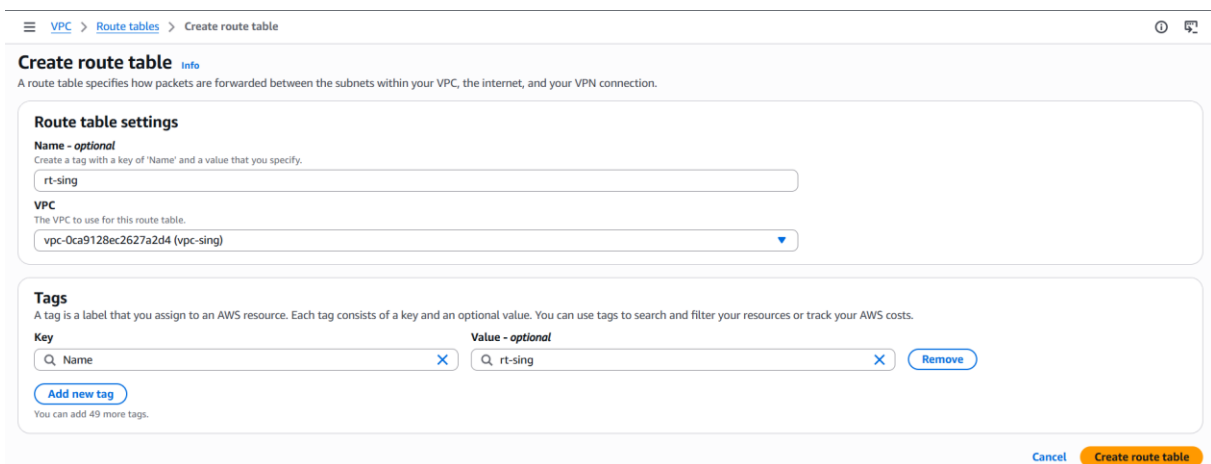
Screenshot: Create Route Table (Singapore)

1. Go to Route Tables
2. Click Create route table



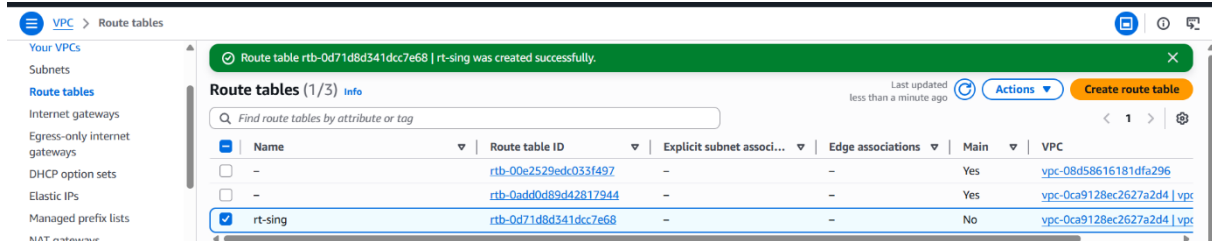
Screenshot: Route Table Details (Singapore)

1. Enter Route table name
2. Select VPC
3. Click Create route table



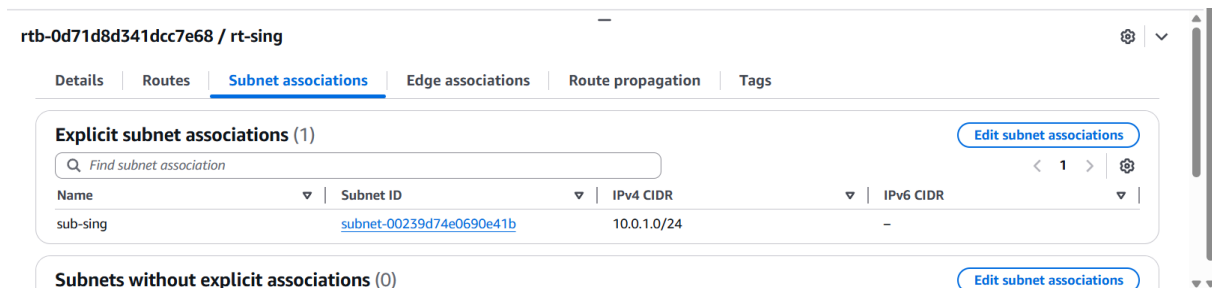
Screenshot: Route Table Created (Singapore)

1. View Route Table
2. Route table created successfully



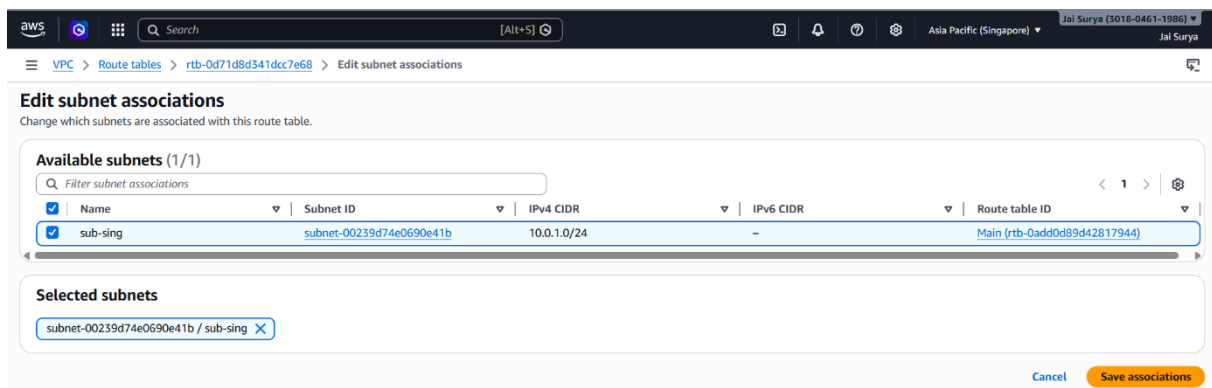
Screenshot: Subnet Association (Singapore)

1. Go to Route Table
2. Click Subnet association → Edit



Screenshot: Edit Subnet Association (Singapore)

1. Click Edit subnet association
2. Select Subnet
3. Click Save



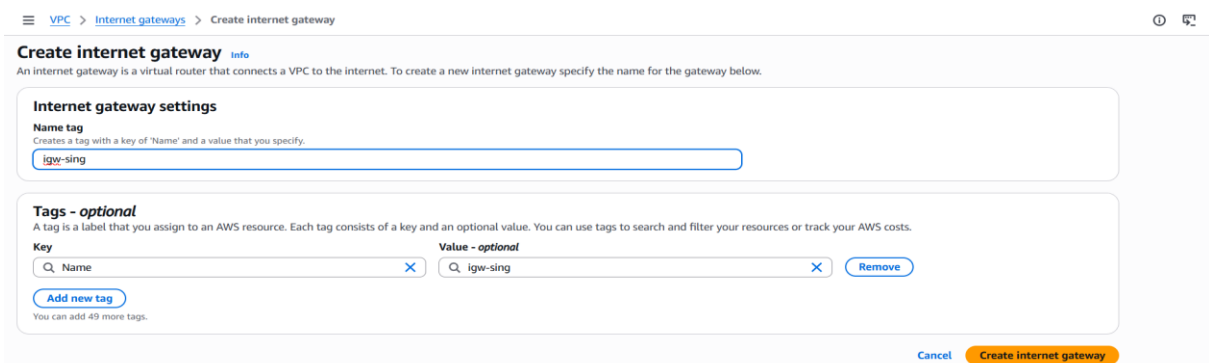
Screenshot: Internet Gateway (Singapore)

1. Go to Internet Gateways
2. Click Create internet gateway



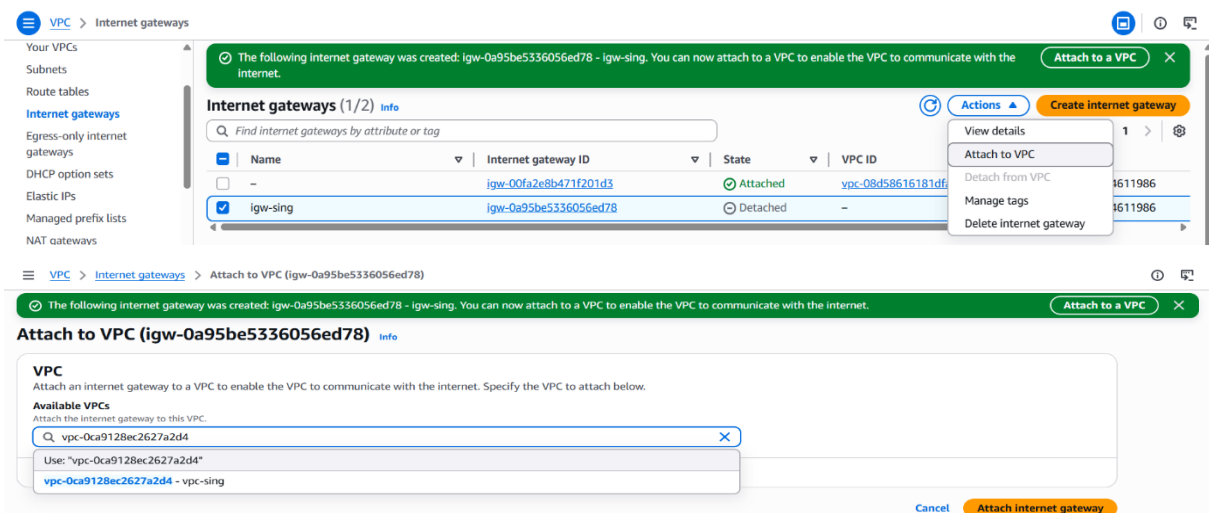
Screenshot: Create Internet Gateway (Singapore)

1. Enter Internet Gateway name
2. Click Create internet gateway



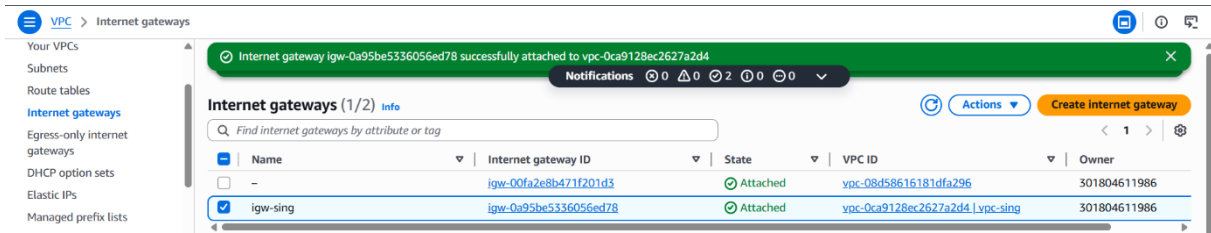
Screenshot: Attach Internet Gateway (Singapore)

1. Internet Gateway created successfully
2. Select Internet Gateway
3. Click Actions → Attach to VPC
4. Select VPC
5. Click Attach



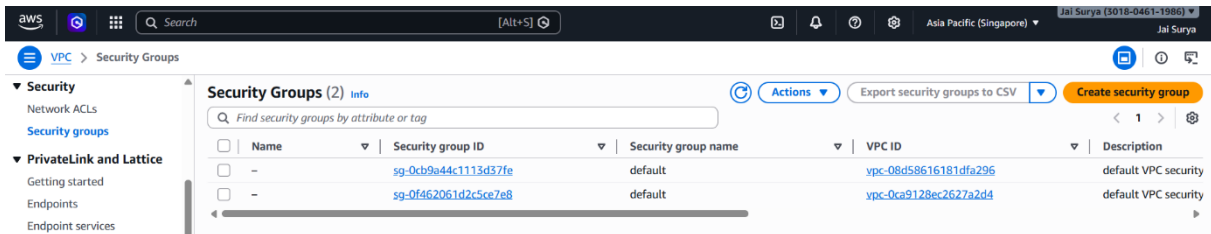
Screenshot: Internet Gateway Attached (Singapore)

1. Internet Gateway attached to VPC successfully



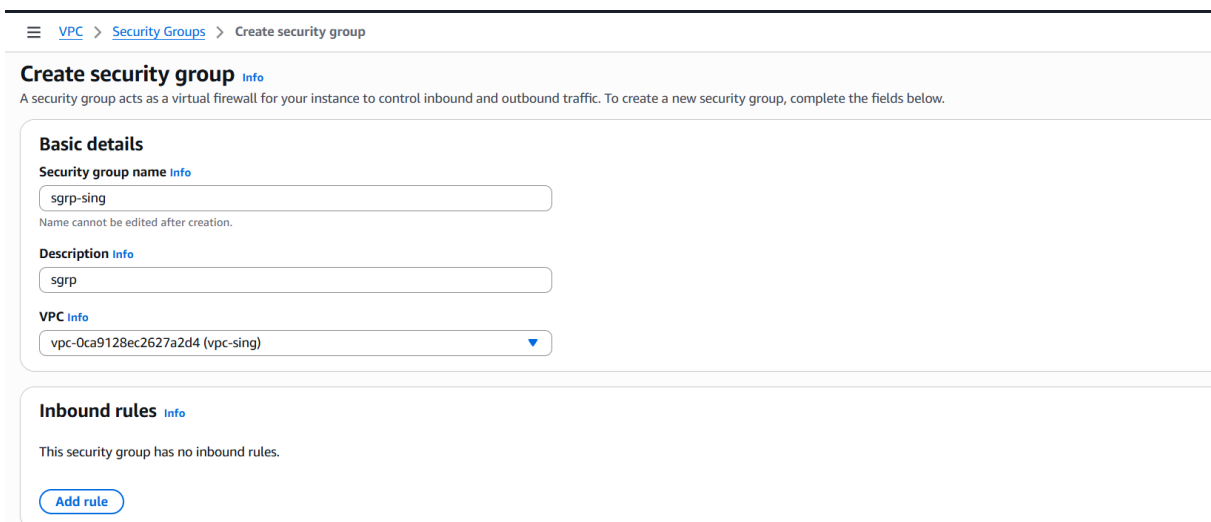
Screenshot: Security Group (Singapore)

1. Go to Security Groups
2. Click Create security group



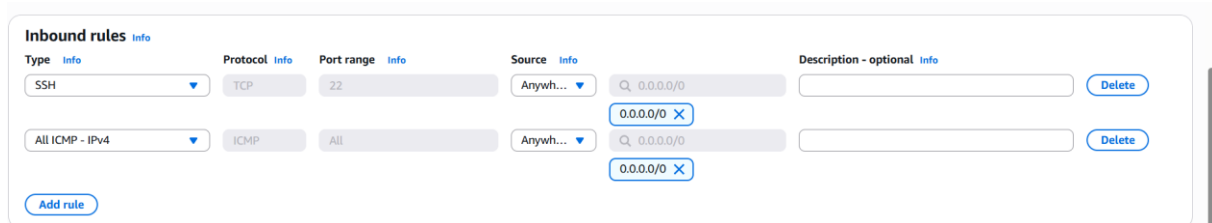
Screenshot: Security Group Details (Singapore)

1. Enter Name
2. Enter Description
3. Select VPC
4. Click Add rule in Inbound rules



Screenshot: Add Inbound Rules (Singapore)

1. Add SSH – Anywhere
2. Add All ICMP IPv4 – Anywhere



Inbound rules [Info](#)

Type [Info](#) Protocol [Info](#) Port range [Info](#) Source [Info](#) Description - optional [Info](#)

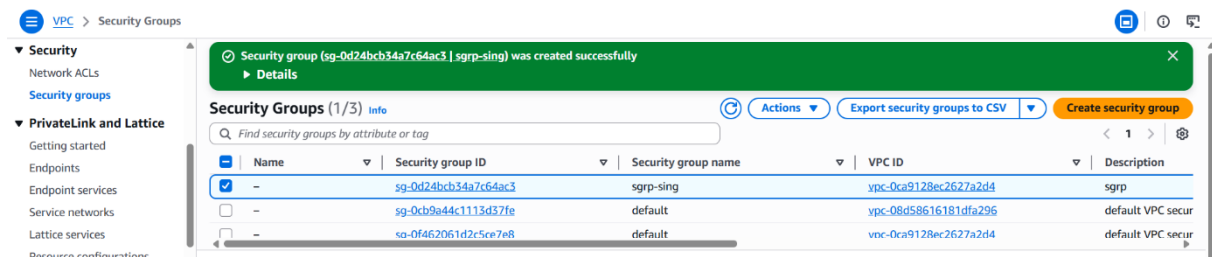
SSH [▼](#) TCP 22 Anywh... [Q](#) 0.0.0.0/0 [Delete](#)

All ICMP - IPv4 [▼](#) ICMP All Anywh... [Q](#) 0.0.0.0/0 [Delete](#)

[Add rule](#)

Screenshot: Security Group Created (Singapore)

1. View Security Group
2. Security group created successfully



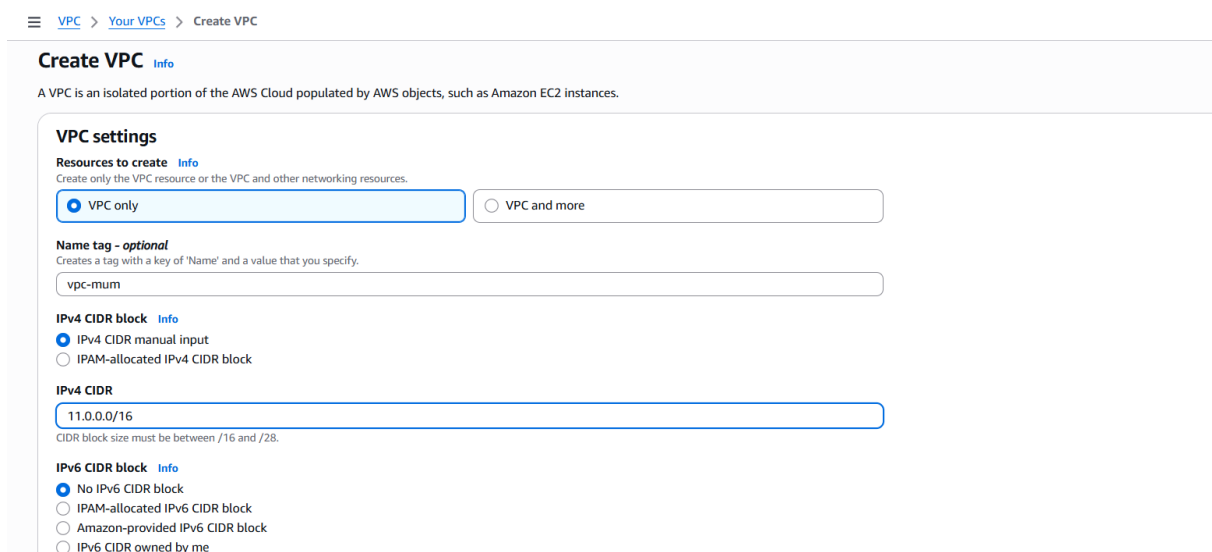
Security Groups (1/3) [Info](#) [Actions](#) [Export security groups to CSV](#) [Create security group](#)

[Find security groups by attribute or tag](#)

Name	Security group ID	Security group name	VPC ID	Description
☒	sg-0d24bcb34a7c64ac3	sgrp-sing	vpc-0ca9128ec2627a2d4	sgrp
☐	sg-0cb9a44c1113d37fe	default	vpc-08d58616181dfa296	default VPC secur
☐	sa-0f462061d2c5ce7e8	default	vpc-0ca9128ec2627a2d4	default VPC secur

Follow the same steps to create VPC, Subnet, Route Table, Internet Gateway, and Security Group in Mumbai region.

Vpc mumbai



Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

vpc-mum

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
11.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

Subnet-mumbai

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

sub-mum

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-s1-az1 (ap-south-1a)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

11.0.0/16

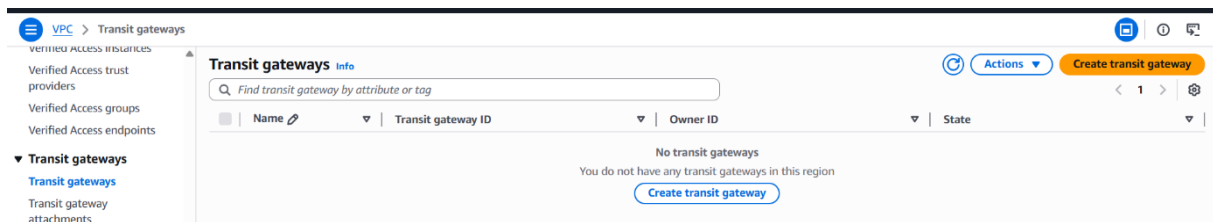
IPv4 subnet CIDR block

11.0.1.0/24

256 IPs

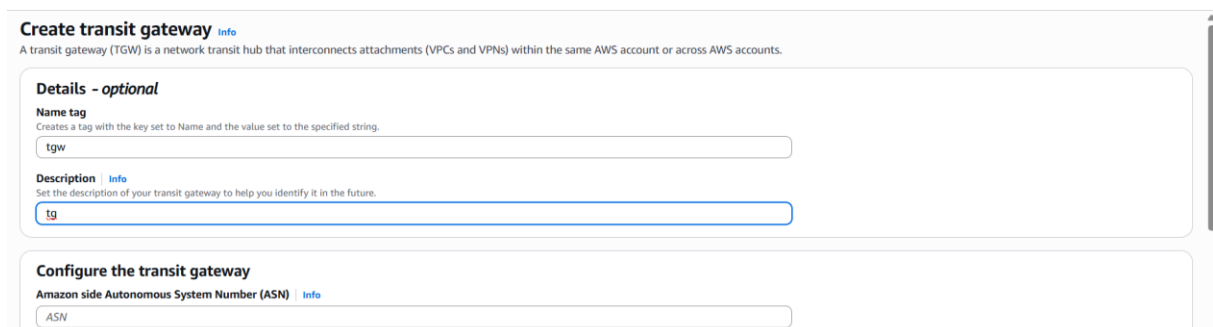
Screenshot: Create Transit Gateway (Singapore & Mumbai)

1. Go to Transit Gateways
2. Click Create transit gateway



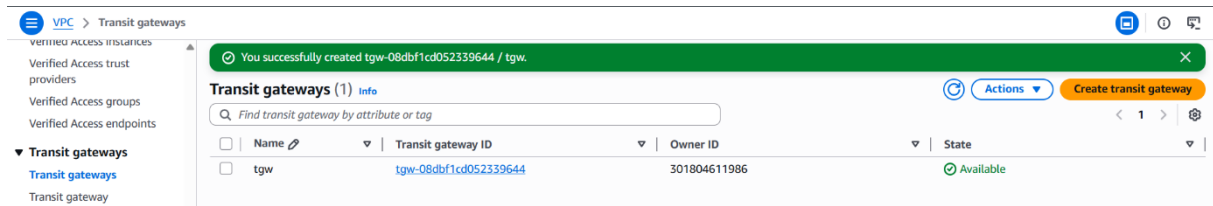
Screenshot: Transit Gateway Details (Singapore & Mumbai)

1. Enter Name and Description
2. Click Create transit gateway



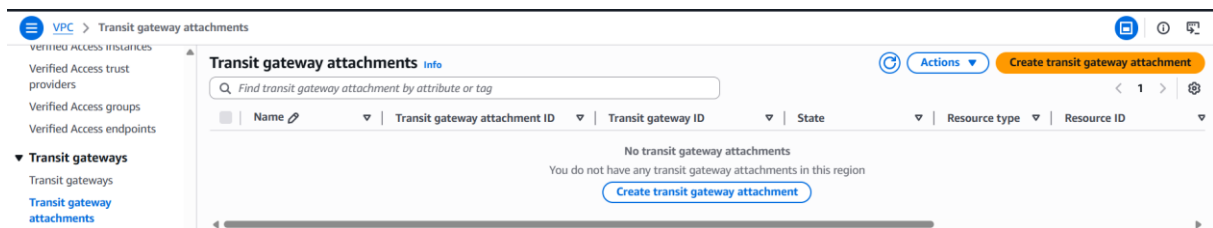
Screenshot: Transit Gateways Created (Singapore & Mumbai)

1. View Transit Gateways
2. Transit Gateways created successfully



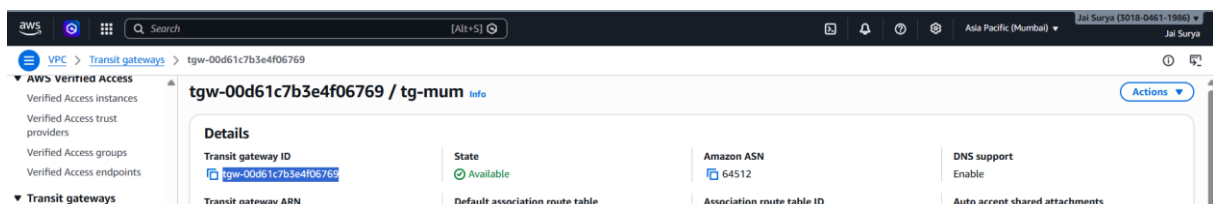
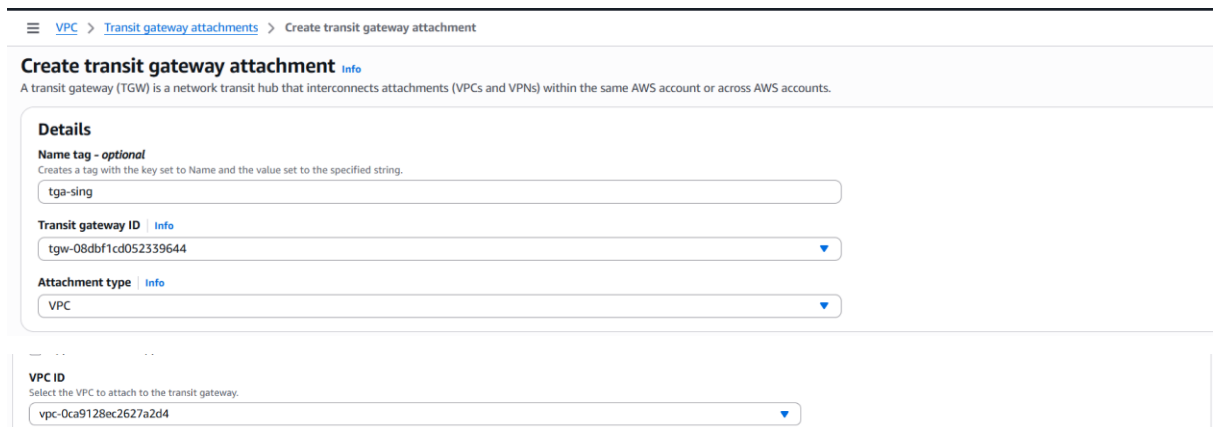
Screenshot: Transit Gateway Attachments (Singapore & Mumbai)

1. Go to Transit Gateway Attachments
2. Click create



Screenshot: Create Transit Gateway Attachment (Singapore & Mumbai)

1. Enter Attachment name
2. Select Transit Gateway ID
3. Select VPC
4. Click Create attachment



Screenshot: Create Peering Attachment (Singapore)

1. Click Create transit gateway attachment
2. Enter Name (Peering)
3. Select Transit Gateway ID
4. Select Attachment type = Peering connection
5. Select My account
6. Select Region = Mumbai
7. Paste Mumbai Transit Gateway ID
8. Click Create attachment

☰ VPC > Transit gateway attachments > Create transit gateway attachment ⓘ

Create transit gateway attachment ⓘ

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional
Creates a tag with the key set to Name and the value set to the specified string.

tga-peering

Transit gateway ID ⓘ
tgw-08dbf1cd052339644

Attachment type ⓘ
Peering connection

Peering connection attachment

Select and configure your peering connection attachment.

Account ⓘ
☒ My account
☐ Other account

Region ⓘ
Asia Pacific (Mumbai) (ap-south-1)

Transit gateway (accepter) ⓘ
tgw-00d61c7b3e4f06769

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Q Name	Q tga-peering	Remove

[Add new tag](#)

You can add up to 49 more tags.

Cancel [Create transit gateway attachment](#)

Screenshot: Peering Attachment Created (Singapore)

1. Transit Gateway attachment created
2. Peering attachment created successfully
3. Status shows Pending acceptance

☰ VPC > Transit gateway attachments ⓘ

Verified Access instances

Verified Access trust providers

Verified Access groups

Verified Access endpoints

▼ Transit gateways

Transit gateways

[Transit gateway attachments](#)

🟢 You successfully created peering attachment tgw-attach-059d0e2ba44d0684d / tga-peering. ⓘ

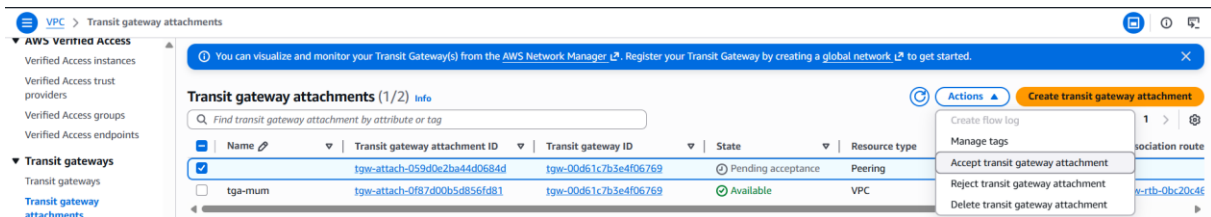
Transit gateway attachments (2) ⓘ [Actions](#) [Create transit gateway attachment](#)

Q Find transit gateway attachment by attribute or tag

☐	Name ⓘ	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
☐	tga-peering	tgw-attach-059d0e2ba44d0684d	tgw-08dbf1cd052339644	⌚ Pending acceptance	Peering	tgw-00d61c7b3e4f06769
☐	tga-sing	tgw-attach-07ae7c3a0c0446e4b	tgw-08dbf1cd052339644	🟢 Available	VPC	vpc-0ca9128ec2627a2d4

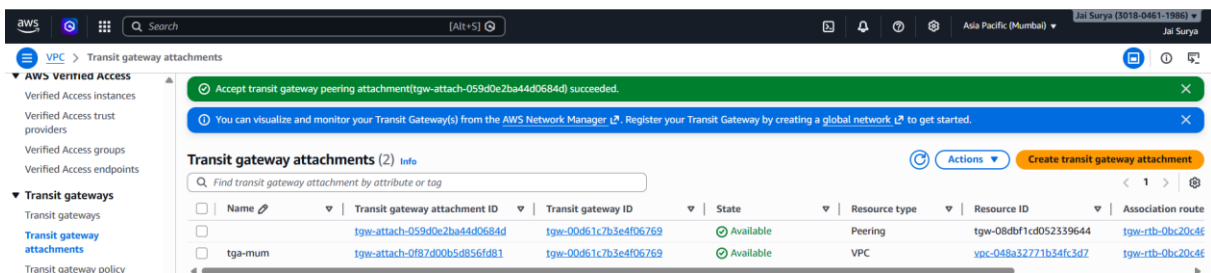
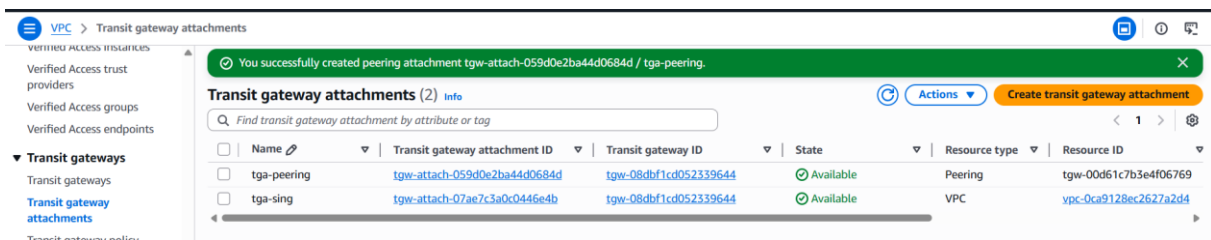
Screenshot: Accept Peering Attachment (Mumbai)

1. Go to Mumbai region
2. Select Peering attachment
3. Click Actions → Accept attachment
4. Peering attachment accepted successfully



Screenshot: Peering Status (Singapore & Mumbai)

1. Check Transit Gateway Attachments in both regions
2. Status shows Available



Screenshot: Create Static Routes (Singapore & Mumbai)

1. Go to Transit Gateway Route Tables
2. Select Transit Gateway Route Table
3. Go to Routes
4. Click Create static routes

Transit gateway route tables (1/1)

Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default
tgw-rtb-012a545bb487e36d4	tgw-rtb-012a545bb487e36d4	tgw-08dbf1cd052339644	Available	Yes	Yes

Transit gateway route tables: tgw-rtb-012a545bb487e36d4

Details | Associations | Propagations | Prefix list references | **Routes** | Tags

Filter routes by CIDR (2)

Exact CIDR
Select a valid IP4 or IPv6 CIDR.
0.0.0.0/0, ::/0

Longest prefix match
Enter a valid IP4 or IPv6 and press enter.
0.0.0.0, ::

Supernet of match
Select a valid IP4 or IPv6 CIDR.
0.0.0.0/0, ::/0

Subnet of match
Select a valid IP4 or IPv6 CIDR.
0.0.0.0/0, ::/0

Routes (1)

CIDR	Attachment ID	Resource ID	Resource type	Route type	Route state
10.0.0.0/16	tgw-attach-07ae7c3a0c0446e4b	vpc-0ca9128ec2627a2d4	VPC	Propagated	Active

Screenshot: Create Static Route (Singapore)

1. Enter CIDR = 11.0.0.0/16 (Mumbai CIDR range)
2. Select Peering attachment
3. Click Create static route

Create static route

Add a static route to your transit gateway route table.

Details

Transit gateway ID
tgw-08dbf1cd052339644

Transit gateway route table ID
tgw-rtb-012a545bb487e36d4

CIDR
11.0.0.0/16

Type
Active

Choose attachment
tgw-attach-059d0e2ba44d0684d

Attachments

- [tgw-attach-059d0e2ba44d0684d \(tga-peering\)](#)
- [tgw-attach-07ae7c3a0c0446e4b \(tga-sing\)](#)

Create static route

Screenshot: Transit Gateway Route Table Updated (Singapore)

1. View Transit Gateway Route Table
2. Static route created successfully

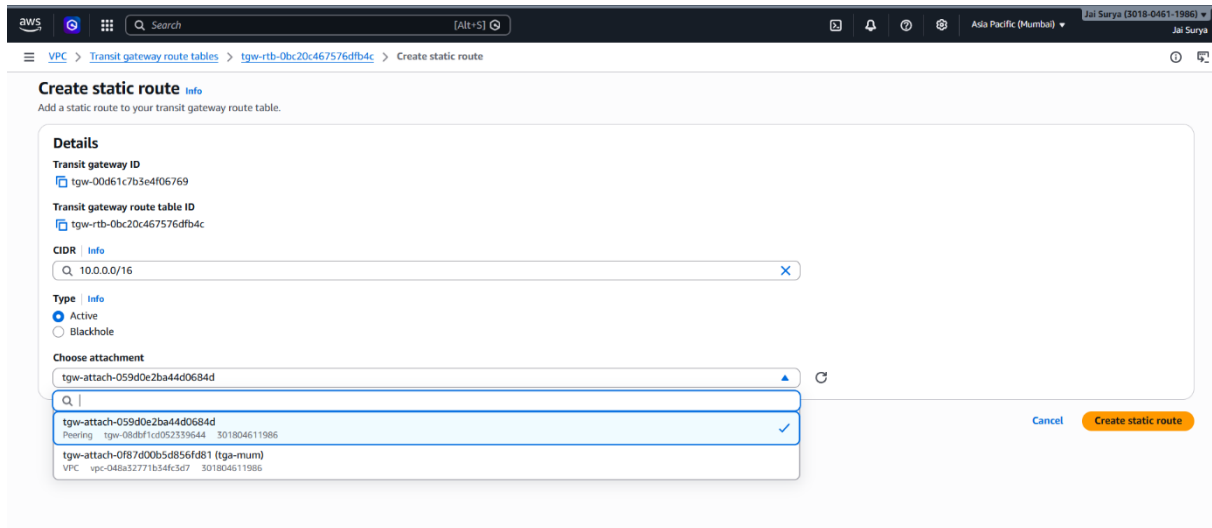
Static route was created successfully. If not immediately visible, the data may still be propagating. Please refresh in a moment.

Transit gateway route tables (1)

Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default
tgw-rtb-012a545bb487e36d4	tgw-rtb-012a545bb487e36d4	tgw-08dbf1cd052339644	Available	Yes	Yes

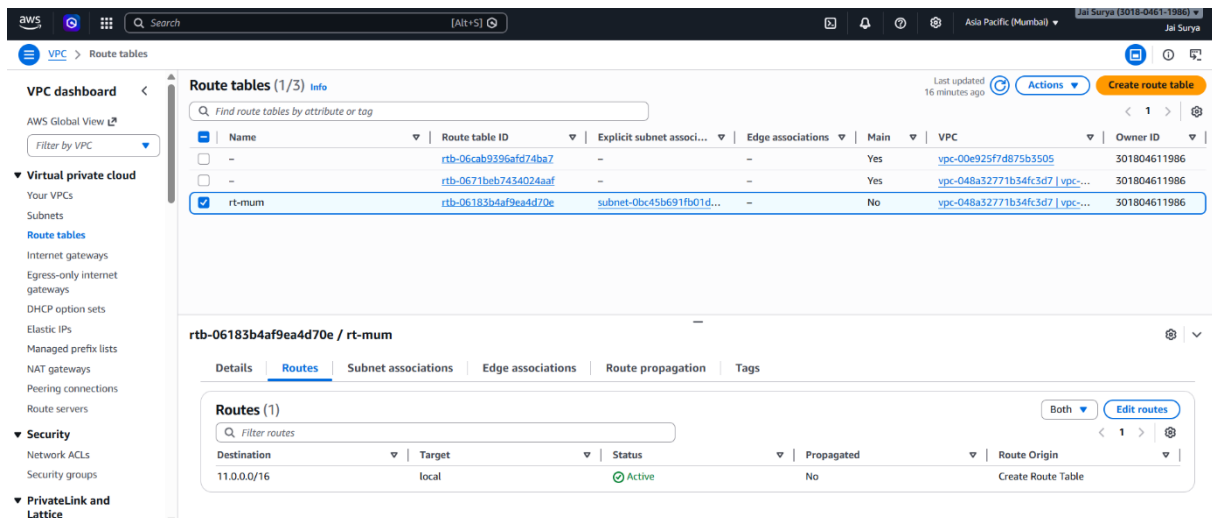
Screenshot: Create Static Route (Mumbai)

1. Enter CIDR = 10.0.0.0/16 (Singapore CIDR range)
2. Select Peering attachment
3. Click Create static route



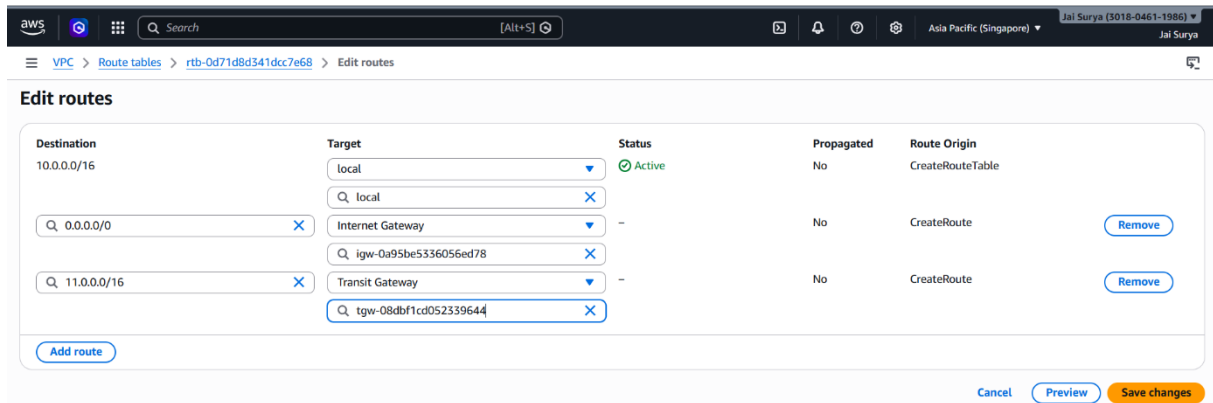
Screenshot: Edit Routes (Singapore & Mumbai)

1. Go to Route Tables
2. Select created Route Table
3. Click Routes → Edit routes



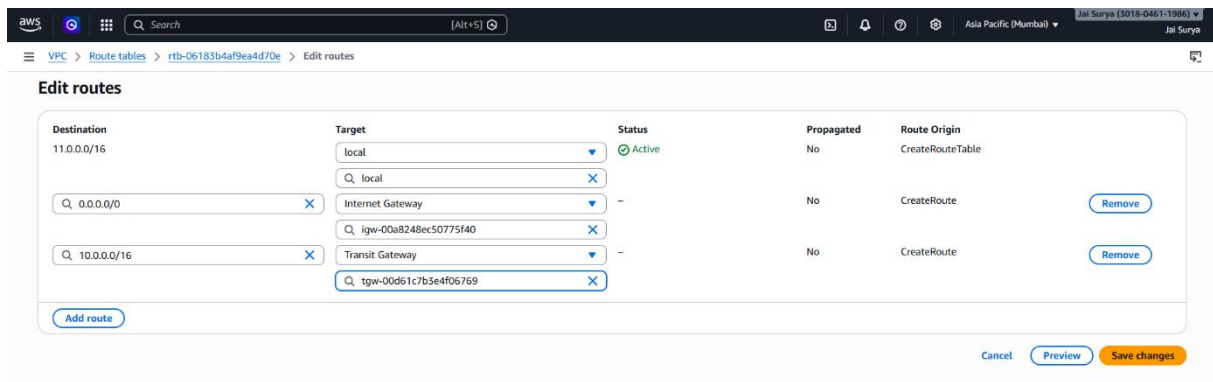
Screenshot: Update Routes (Singapore)

1. Add 0.0.0.0/0 → Choose Internet Gateway → Select IGW ID
2. Add 11.0.0.0/16 → Choose Transit Gateway → Select Transit Gateway ID
3. Click Save



Screenshot: Update Routes (Mumbai)

1. Add 0.0.0.0/0 → Choose Internet Gateway → Select IGW ID
2. Add 10.0.0.0/16 → Choose Transit Gateway → Select Transit Gateway ID
3. Click Save



Screenshot: Create Linux Instance (Singapore & Mumbai)

1. Go to EC2
2. Click Launch Instance
3. Select Linux AMI
4. Create instances in both regions (S and M)
5. Click Launch

aws

Search

[Alt+S]

Asia Pacific (Singapore)

Jai Surya (3018-0461-1986)

Jai Surya

EC2

Instances

Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

linux-sing

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

RecentsQuick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)

ami-0be9c9f67c8dabd6

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

singapore

Create new key pair

aws

Search

[Alt+S]

Asia Pacific (Singapore)

Jai Surya (3018-0461-1986)

Jai Surya

EC2

Instances

Launch an instance

Network settings

VPC - required

vpc-0ca9128ec2627a2d4 (vpc-sing)

10.0.0.0/16

Subnet

subnet-00239d74e0690e41b

sub-sing

VPC: vpc-0ca9128ec2627a2d4 Owner: 301804611986 Availability Zone: ap-southeast-1a (apse1-a-z2) Zone type: Availability Zone IP addresses available: 250 CIDR: 10.0.1.0/24

Auto-assign public IP

Enable

Firewall (security groups)

Create security group

Select existing security group

Common security groups

Select security groups

sgp-sing sg-0d24bc34a7c64ac3

VPC: vpc-0ca9128ec2627a2d4

Cancel

Launch instance

Preview code

aws

Search

[Alt+S]

Ask Amazon Q

Asia Pacific (Mumbai)

Jai Surya (3018-0461-1986)

Jai Surya

EC2

Instances

Launch an instance

Network settings

VPC - required

vpc-048a32771b34fc3d7 (vpc-mum)

11.0.0.0/16

Subnet

subnet-0bc45b691fb01daeb

sub-mum

VPC: vpc-048a32771b34fc3d7 Owner: 301804611986 Availability Zone: ap-south-1a (aps1-a-z1) Zone type: Availability Zone IP addresses available: 250 CIDR: 11.0.1.0/24

Auto-assign public IP

Enable

Firewall (security groups)

Create security group

Select existing security group

Common security groups

Select security groups

sgrp-mum sg-0b991732715b8a404

VPC: vpc-048a32771b34fc3d7

Advanced network configuration

Cancel

Launch instance

Preview code

Screenshot: Connectivity Test (Singapore)

1. Open Linux Terminal (Singapore)
2. Run ping using Mumbai instance Private IP
3. Check response for connectivity



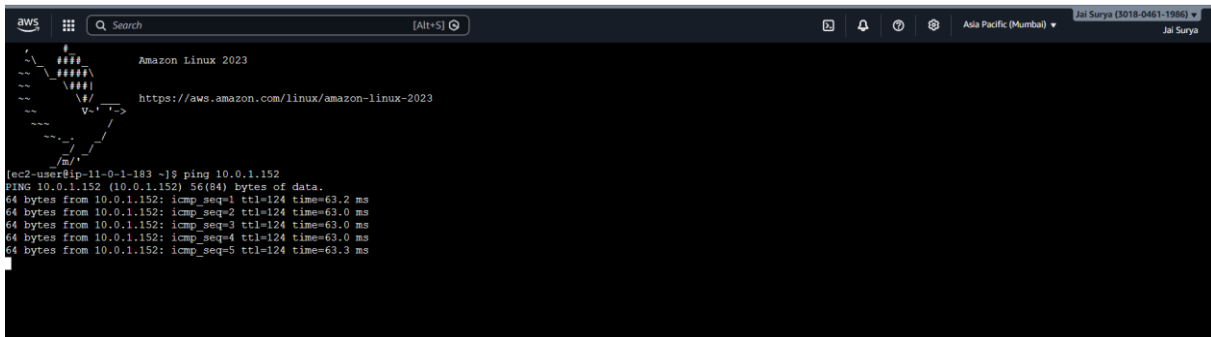
The screenshot shows an AWS Linux terminal window. The title bar indicates the region is 'Asia Pacific (Singapore)'. The terminal displays the Amazon Linux 2023 logo and a URL: <https://aws.amazon.com/linux/amazon-linux-2023>. The user is logged in as 'ec2-user' with IP 'ip-10-0-1-152'. They run the command 'ping 11.0.1.183'. The output shows successful connectivity with 7 ping attempts, each receiving 64 bytes from 11.0.1.183 with a TTL of 124 and times ranging from 63.0 ms to 67.2 ms.

```
aws [Alt+S] Jai Surya (3018-0461-1986) Jai Surya
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-1-152 ~]$ ping 11.0.1.183
PING 11.0.1.183 (11.0.1.183) 56(84) bytes of data.
64 bytes from 11.0.1.183: icmp_seq=1 ttl=124 time=67.2 ms
64 bytes from 11.0.1.183: icmp_seq=2 ttl=124 time=63.1 ms
64 bytes from 11.0.1.183: icmp_seq=3 ttl=124 time=64.0 ms
64 bytes from 11.0.1.183: icmp_seq=4 ttl=124 time=63.0 ms
64 bytes from 11.0.1.183: icmp_seq=5 ttl=124 time=63.1 ms
64 bytes from 11.0.1.183: icmp_seq=6 ttl=124 time=63.0 ms
64 bytes from 11.0.1.183: icmp_seq=7 ttl=124 time=63.0 ms
```

Screenshot: Connectivity Test (Mumbai)

1. Open Linux Terminal (Mumbai)
2. Run ping using Singapore instance Private IP
3. Check response for connectivity



The screenshot shows an AWS Linux terminal window. The title bar indicates the region is 'Asia Pacific (Mumbai)'. The terminal displays the Amazon Linux 2023 logo and a URL: <https://aws.amazon.com/linux/amazon-linux-2023>. The user is logged in as 'ec2-user' with IP 'ip-11-0-1-183'. They run the command 'ping 10.0.1.152'. The output shows successful connectivity with 5 ping attempts, each receiving 64 bytes from 10.0.1.152 with a TTL of 124 and times ranging from 63.0 ms to 63.3 ms.

```
aws [Alt+S] Jai Surya (3018-0461-1986) Jai Surya
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-11-0-1-183 ~]$ ping 10.0.1.152
PING 10.0.1.152 (10.0.1.152) 56(84) bytes of data.
64 bytes from 10.0.1.152: icmp_seq=1 ttl=124 time=63.2 ms
64 bytes from 10.0.1.152: icmp_seq=2 ttl=124 time=63.0 ms
64 bytes from 10.0.1.152: icmp_seq=3 ttl=124 time=63.0 ms
64 bytes from 10.0.1.152: icmp_seq=4 ttl=124 time=63.0 ms
64 bytes from 10.0.1.152: icmp_seq=5 ttl=124 time=63.3 ms
```